

Aggregation in SQL

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Assignment Name: Aggregation in SQL

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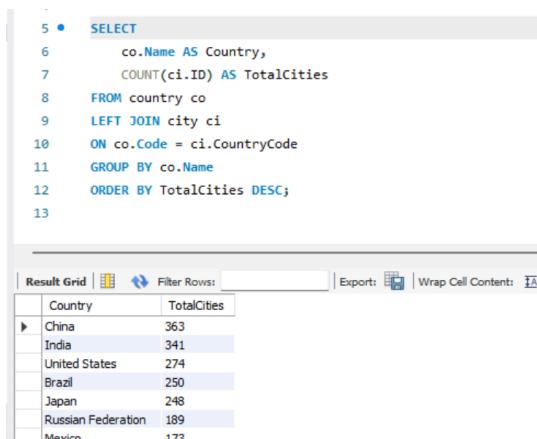
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Q1. Count how many cities are there in each country?

Answer:

USE world;

```
SELECT
    co.Name AS Country,
    COUNT(ci.ID) AS TotalCities
FROM country co
LEFT JOIN city ci
ON co.Code = ci.CountryCode
GROUP BY co.Name
ORDER BY TotalCities DESC;
```



The screenshot shows a SQL query editor with the following query:

```
5 • SELECT
6     co.Name AS Country,
7     COUNT(ci.ID) AS TotalCities
8 FROM country co
9 LEFT JOIN city ci
10 ON co.Code = ci.CountryCode
11 GROUP BY co.Name
12 ORDER BY TotalCities DESC;
13
```

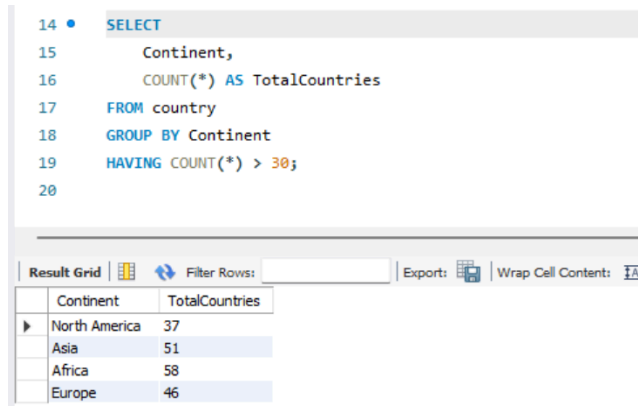
Below the query editor is a 'Result Grid' showing the results of the query. The grid has two columns: 'Country' and 'TotalCities'. The results are ordered by 'TotalCities' in descending order.

Country	TotalCities
China	363
India	341
United States	274
Brazil	250
Japan	248
Russian Federation	189
Mexico	173

Q2. Display all continents having more than 30 countries.

Answer:

```
SELECT
    Continent,
    COUNT(*) AS TotalCountries
FROM country
GROUP BY Continent
HAVING COUNT(*) > 30;
```



The screenshot shows a SQL query editor with the following query:

```
14 • SELECT
15     Continent,
16     COUNT(*) AS TotalCountries
17 FROM country
18 GROUP BY Continent
19 HAVING COUNT(*) > 30;
20
```

Below the query editor is a 'Result Grid' showing the results of the query. The grid has two columns: 'Continent' and 'TotalCountries'. The results are filtered to show only continents with more than 30 countries.

Continent	TotalCountries
North America	37
Asia	51
Africa	58
Europe	46

Q3. List regions whose total population exceeds 200 million.

Answer:

```
SELECT
    Region,
    SUM(Population) AS TotalPopulation
FROM country
GROUP BY Region
HAVING SUM(Population) > 200000000;
```

21 • **SELECT**
 22 Region,
 23 SUM(Population) AS TotalPopulation
 24 **FROM** country
 25 **GROUP BY** Region
 26 **HAVING** SUM(Population) > 200000000;
 27

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [FA](#)

Region	TotalPopulation
Southern and Central Asia	1490776000
South America	345780000
Eastern Africa	246999000
Western Africa	221672000
Eastern Europe	307026000
North America	309632000
Southeast Asia	518541000

Q4. Find the top 5 continents by average GNP per country.

Answer:

```
SELECT
    Continent,
    AVG(GNP) AS AverageGNP
FROM country
GROUP BY Continent
ORDER BY AverageGNP DESC
LIMIT 5;
```

28 • **SELECT**
 29 Continent,
 30 AVG(GNP) AS AverageGNP
 31 **FROM** country
 32 **GROUP BY** Continent
 33 **ORDER BY** AverageGNP DESC
 34 **LIMIT** 5;
 35

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [FA](#)

Continent	AverageGNP
North America	261854.789189
Europe	206497.065217
Asia	150105.725490
South America	107991.000000
Oceania	14991.953571

Q5. Find the total number of official languages spoken in each continent.

Answer:

```
SELECT
    c.Continent,
    COUNT(cl.Language) AS OfficialLanguages
FROM country c
JOIN countrylanguage cl
ON c.Code = cl.CountryCode
WHERE cl.IsOfficial = 'T'
GROUP BY c.Continent;
```

36 • **SELECT**
 37 c.Continent,
 38 COUNT(cl.Language) AS OfficialLanguages
 39 **FROM** country c
 40 **JOIN** countrylanguage cl
 41 **ON** c.Code = cl.CountryCode
 42 **WHERE** cl.IsOfficial = 'T'
 43 **GROUP BY** c.Continent;
 44
 45

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [FA](#)

Continent	OfficialLanguages
North America	35
Asia	57
Europe	59
South America	16
Oceania	33
Africa	38

Q6. Find the maximum and minimum GNP for each continent.

Answer:

```
SELECT
    Continent,
    MAX(GNP) AS MaximumGNP,
    MIN(GNP) AS MinimumGNP
FROM country
GROUP BY Continent;
```

```
45 • SELECT
46     Continent,
47     MAX(GNP) AS MaximumGNP,
48     MIN(GNP) AS MinimumGNP
49 FROM country
50 GROUP BY Continent;
51
52
```

Continent	MaximumGNP	MinimumGNP
North America	8510700.00	0.00
Asia	3787042.00	0.00
Africa	116729.00	0.00
Europe	2133367.00	0.00
South America	776739.00	0.00
Oceania	351182.00	0.00
Antarctica	0.00	0.00

Q7. Find the country with the highest average city population.

Answer:

```
SELECT
    co.Name AS Country,
    AVG(ci.Population) AS AverageCityPopulation
FROM country co
JOIN city ci
ON co.Code = ci.CountryCode
GROUP BY co.Name
ORDER BY AverageCityPopulation DESC
LIMIT 1;
```

```
52 • SELECT
53     co.Name AS Country,
54     AVG(ci.Population) AS AverageCityPopulation
55 FROM country co
56 JOIN city ci
57 ON co.Code = ci.CountryCode
58 GROUP BY co.Name
59 ORDER BY AverageCityPopulation DESC
60 LIMIT 1;
61
```

Country	AverageCityPopulation
Singapore	4017733.0000

Q8. List continents where the average city population is greater than 200,000.

Answer:

```
SELECT
    co.Continent,
    AVG(ci.Population) AS AverageCityPopulation
FROM country co
JOIN city ci
ON co.Code = ci.CountryCode
GROUP BY co.Continent
HAVING AVG(ci.Population) > 200000;
```

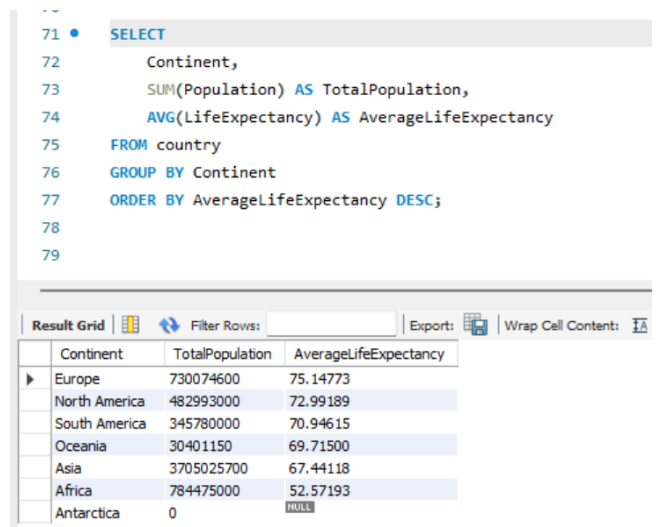
```
62 • SELECT
63     co.Continent,
64     AVG(ci.Population) AS AverageCityPopulation
65 FROM country co
66 JOIN city ci
67 ON co.Code = ci.CountryCode
68 GROUP BY co.Continent
69 HAVING AVG(ci.Population) > 200000;
70
71
```

Continent	AverageCityPopulation
North America	289587.5749
Asia	395019.3109
Africa	371143.6585
Europe	287684.6766
South America	366037.9979
Oceania	252475.4364

Q9. Find the total population and average life expectancy for each continent, ordered by average life expectancy descending.

Answer:

```
SELECT
    Continent,
    SUM(Population) AS TotalPopulation,
    AVG(LifeExpectancy) AS AverageLifeExpectancy
FROM country
GROUP BY Continent
ORDER BY AverageLifeExpectancy DESC;
```



The screenshot shows a SQL query editor with the following query:

```
71 • SELECT
72     Continent,
73     SUM(Population) AS TotalPopulation,
74     AVG(LifeExpectancy) AS AverageLifeExpectancy
75 FROM country
76 GROUP BY Continent
77 ORDER BY AverageLifeExpectancy DESC;
78
79
```

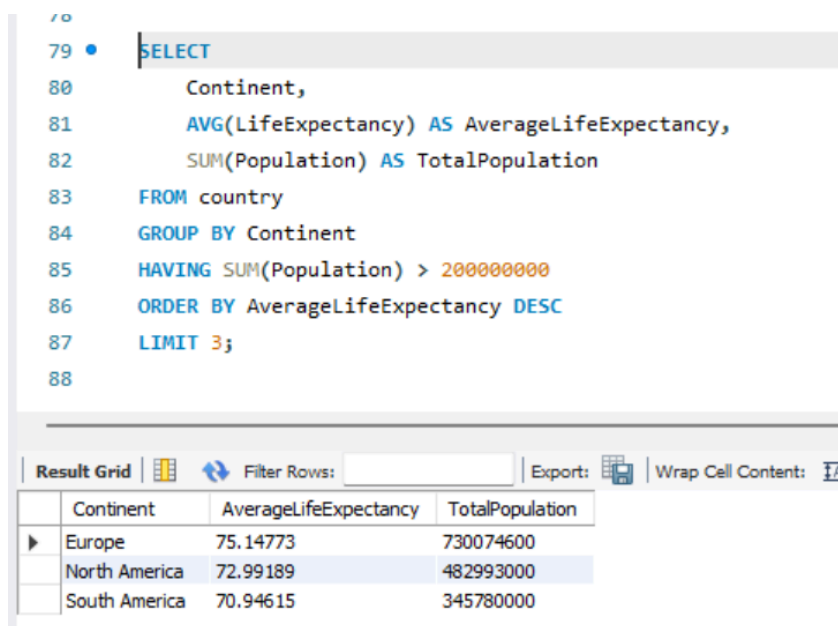
Below the query editor is a 'Result Grid' showing the results of the query:

Continent	TotalPopulation	AverageLifeExpectancy
Europe	730074600	75.14773
North America	482993000	72.99189
South America	345780000	70.94615
Oceania	30401150	69.71500
Asia	3705025700	67.44118
Africa	784475000	52.57193
Antarctica	0	NULL

Q10. Find the top 3 continents with the highest average life expectancy, but only include those where the total population is over 200 million.

Answer:

```
SELECT
    Continent,
    AVG(LifeExpectancy) AS AverageLifeExpectancy,
    SUM(Population) AS TotalPopulation
FROM country
GROUP BY Continent
HAVING SUM(Population) > 200000000
ORDER BY AverageLifeExpectancy DESC
LIMIT 3;
```



The screenshot shows a SQL query editor with the following query:

```
79 • SELECT
80     Continent,
81     AVG(LifeExpectancy) AS AverageLifeExpectancy,
82     SUM(Population) AS TotalPopulation
83 FROM country
84 GROUP BY Continent
85 HAVING SUM(Population) > 200000000
86 ORDER BY AverageLifeExpectancy DESC
87 LIMIT 3;
88
```

Below the query editor is a 'Result Grid' showing the results of the query:

Continent	AverageLifeExpectancy	TotalPopulation
Europe	75.14773	730074600
North America	72.99189	482993000
South America	70.94615	345780000